Schwarzschild Geometry Without Singularity: A Phase-Parameter Derivation

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Abstract

We derive the Schwarzschild solution to the Einstein vacuum equations using a phase parameter as the fundamental radial variable, with the areal radius appearing only as a derived quantity. The resulting metric reproduces Schwarzschild geometry on the entire physical domain and predicts identical observational consequences. The curvature singularity at $r = 0$ does not appear in the derivation, because the locus $r = 0$ lies outside the domain of the generating phase parameter. We conclude that the Schwarzschild singularity is an artefact of treating the radial coordinate as primitive, not a property of the geometry itself. The argument requires no modification of general relativity and no exotic matter content.

1 Introduction

The exterior Schwarzschild solution, derived in 1916, is the unique static spherically-symmetric vacuum solution of Einstein's field equations. In its standard form,

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2\,, \tag{1}$$

with $d\Omega^2 = d\varphi^2 + \sin^2\varphi\, d\psi^2$, the metric exhibits two distinguished loci. The first, at $r = 2GM$, is a coordinate singularity removable by transition to Kruskal-Szekeres coordinates. The second, at $r = 0$, is a curvature singularity at which the Kretschmann scalar

$$K \equiv R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} = \frac{48\, G^2 M^2}{r^6} \tag{2}$$

diverges.

A substantial fraction of theoretical physics over the past century has been organised around the second locus: singularity theorems, cosmic censorship, black hole interior structure, the information paradox, holographic dualities, and proposed completions of general relativity intended to resolve the divergence. The present note observes that the divergence appears only when the radial direction is parameterised by a coordinate r that ranges over the entire half-line $[0, \infty)$. We exhibit a parameterisation in which the radial structure is generated by a phase parameter whose image is strictly bounded away from zero, derive the Einstein vacuum equations in that parameterisation, and observe that the resulting solution reproduces Schwarzschild geometry on the entire physical domain while the singular locus simply does not exist.

2 The Phase Parameterisation

Definition 1 (Phase parameter and generative function)**.** Let $\theta \in [0, \infty)$ be a real parameter. Define the generative function

$$\rho \colon [0, \infty) \to [1, \infty), \qquad \rho(\theta) \;=\; 1 + \frac{2\theta}{\Theta_{\max}}\,, \tag{3}$$

where $\Theta_{\max} > 0$ is a fixed constant with the dimensions of θ, representing the phase advance per revolution.

The function ρ has the following elementary properties.

Lemma 2 (Properties of ρ)**.** *The function $\rho \colon [0, \infty) \to [1, \infty)$ defined by (3) satisfies:*

(i) $\rho(0) = 1$*;*

(ii) $\rho'(\theta) = 2/\Theta_{\max} > 0$ *for all* $\theta \in [0, \infty)$*, so ρ is strictly monotonically increasing;*

(iii) $\lim_{\theta \to \infty} \rho(\theta) = +\infty$*;*

(iv) ρ *is* C^∞ *on* $[0, \infty)$*;*

(v) the image of ρ is the half-line $[1, \infty)$*, and in particular* $\rho(\theta) \geq 1 > 0$ *for every* $\theta \in [0, \infty)$*.*

Proof. All claims are immediate from (3). For (v), monotonicity together with $\rho(0) = 1$ and $\rho(\theta) \to \infty$ as $\theta \to \infty$ gives the image as $[1, \infty)$. □

Remark 3. Property (v) of Lemma 2 is the structurally essential feature of the parameterisation. The value $\rho = 0$ is not attained for any θ in the domain. Setting $\rho(\theta_*) = 0$ in (3) formally yields $\theta_* = -\Theta_{\max}/2 < 0$, which lies outside $[0, \infty)$. The locus $\rho = 0$ is therefore not in the image of any phase parameter value; it corresponds to a hypothetical *negative-phase* value that the parameter does not admit.

Definition 4 (Areal radius)**.** On any static spherically symmetric spacetime, the areal radius at a given radial parameter value is defined by

$$\rho \;=\; \sqrt{\frac{A}{4\pi}}\,, \tag{4}$$

where A is the surface area of the 2-sphere of orbits of the rotation group at that parameter value.

In what follows we identify the function $\rho(\theta)$ of Definition 1 with the areal radius of Definition 4. Geometrically: the 2-sphere at phase parameter θ has surface area $4\pi\rho(\theta)^2$, and this surface area grows monotonically and without bound as θ advances.

3 Metric Ansatz and Einstein Equations

3.1 The static spherically symmetric ansatz

Let $\mathcal{M}$ be a four-dimensional Lorentzian manifold equipped with coordinates $(t, \theta, \varphi, \psi)$, with $t \in \mathbb{R}$ a time coordinate, $\theta \in [0, \infty)$ the phase parameter of Definition 1, and $(\varphi, \psi) \in [0, \pi] \times [0, 2\pi)$ standard coordinates on the 2-sphere. The most general static, spherically symmetric line element in this chart is

$$ds^2 \;=\; -F(\theta)\, dt^2 \;+\; G(\theta)\, d\theta^2 \;+\; \rho(\theta)^2\, d\Omega^2\,, \tag{5}$$

where $F, G \colon [0, \infty) \to (0, \infty)$ are smooth positive functions to be determined, and $d\Omega^2$ is the round metric on the unit 2-sphere.

3.2 Christoffel symbols

We compute the nonzero Christoffel symbols of the metric (5). Adopting the index convention $(x^0, x^1, x^2, x^3) = (t, \theta, \varphi, \psi)$, the nonzero connection coefficients are:

$$\Gamma^0_{01} = \Gamma^0_{10} = \frac{F'(\theta)}{2F(\theta)}\,, \tag{6}$$

$$\Gamma^1_{00} = \frac{F'(\theta)}{2G(\theta)}\,, \tag{7}$$

$$\Gamma^1_{11} = \frac{G'(\theta)}{2G(\theta)}\,, \tag{8}$$

$$\Gamma^1_{22} = -\frac{\rho(\theta)\rho'(\theta)}{G(\theta)}\,, \tag{9}$$

$$\Gamma^1_{33} = -\frac{\rho(\theta)\rho'(\theta)\sin^2\varphi}{G(\theta)}\,, \tag{10}$$

$$\Gamma^2_{12} = \Gamma^2_{21} = \Gamma^3_{13} = \Gamma^3_{31} = \frac{\rho'(\theta)}{\rho(\theta)}\,, \tag{11}$$

$$\Gamma^2_{33} = -\sin\varphi\cos\varphi\,, \tag{12}$$

$$\Gamma^3_{23} = \Gamma^3_{32} = \cot\varphi\,. \tag{13}$$

Primes denote differentiation with respect to θ.

3.3 Ricci tensor

The nonzero components of the Ricci tensor are obtained from

$$R_{\mu\nu} = \partial_\rho\Gamma^\rho_{\mu\nu} - \partial_\nu\Gamma^\rho_{\mu\rho} + \Gamma^\rho_{\rho\sigma}\Gamma^\sigma_{\mu\nu} - \Gamma^\rho_{\nu\sigma}\Gamma^\sigma_{\mu\rho}\,. \tag{14}$$

Direct computation yields:

$$R_{00} = \frac{F''}{2G} - \frac{F'}{4G}\left(\frac{F'}{F} + \frac{G'}{G}\right) + \frac{F'\rho'}{G\rho}\,, \tag{15}$$

$$R_{11} = -\frac{F''}{2F} + \frac{F'}{4F}\left(\frac{F'}{F} + \frac{G'}{G}\right) + \frac{G'\rho'}{G\rho} - \frac{2\rho''}{\rho}\,, \tag{16}$$

$$R_{22} = 1 - \frac{\rho\rho''}{G} - \frac{(\rho')^2}{G} - \frac{\rho\rho'}{2G}\left(\frac{F'}{F} - \frac{G'}{G}\right)\,, \tag{17}$$

$$R_{33} = \sin^2\varphi \cdot R_{22}\,. \tag{18}$$

3.4 Vacuum equations

The Einstein vacuum equations $R_{\mu\nu} = 0$ reduce, in this ansatz, to the three independent ordinary differential equations

$$R_{00} = 0\,, \tag{19}$$

$$R_{11} = 0\,, \tag{20}$$

$$R_{22} = 0\,. \tag{21}$$

The equation $R_{33} = 0$ follows from (21) by the proportionality (18).

4 Solution of the Vacuum Equations

4.1 First integral

Combine $G \cdot R_{00} + F \cdot R_{11} = 0$. After cancellation of the second-derivative terms:

$$\frac{F'\rho'}{\rho} + \frac{FG'\rho'}{G\rho} - 2F\frac{\rho''}{\rho} = 0\,. \tag{22}$$

For the linear $\rho(\theta) = 1 + 2\theta/\Theta_{\max}$, $\rho'' \equiv 0$, so the third term vanishes. The remaining equation is

$$\frac{F'}{F} + \frac{G'}{G} = 0\,, \tag{23}$$

which integrates immediately to

$$F(\theta)\,G(\theta) = (\rho'(\theta))^2 \cdot C_1 \tag{24}$$

for a constant $C_1 > 0$. The factor $(\rho')^2$ is included for later convenience; absorbing it into C_1 would not change the geometry.

Without loss of generality we set $C_1 = 1$, which corresponds to choosing units in which t measures proper time at spatial infinity. This gives

$$G(\theta) = \frac{(\rho'(\theta))^2}{F(\theta)}\,. \tag{25}$$

4.2 Solving for F

Substitute (25) into (21). After expansion:

$$1 - \frac{\rho\rho''F}{(\rho')^2} - F - \rho F' = 0\,. \tag{26}$$

For linear ρ, $\rho'' = 0$, so

$$1 - F(\theta) - \rho(\theta) \cdot F'(\theta) = 0\,. \tag{27}$$

This is a first-order linear ODE for F as a function of θ. Treating ρ as the independent variable via the substitution $u = \rho(\theta)$, with $du = \rho'\,d\theta$:

$$1 - F(u) - u\frac{dF}{du} = 0\,, \qquad u \in [1, \infty)\,. \tag{28}$$

Rearranging:

$$u\frac{dF}{du} + F(u) = 1\,. \tag{29}$$

The left side is $d(uF)/du$. Integrating:

$$uF(u) = u + C_2\,, \tag{30}$$

where C_2 is a constant of integration. Solving for F:

$$F(u) = 1 + \frac{C_2}{u}\,. \tag{31}$$

Identification of the integration constant with the Schwarzschild mass parameter is fixed by the Newtonian limit. In the weak-field regime at large ρ, the time-time component of the metric must reproduce the Newtonian gravitational potential:

$$F(\rho) \;\to\; 1 - \frac{2GM}{\rho} \qquad \text{as } \rho \to \infty\,, \tag{32}$$

which gives $C_2 = -2GM$.

4.3 The metric

Substituting back into (5), the line element on the manifold is

$$\boxed{ds^2 \;=\; -\left(1-\frac{2GM}{\rho(\theta)}\right)dt^2 \;+\; \left(\frac{2}{\Theta_{\max}}\right)^2\left(1-\frac{2GM}{\rho(\theta)}\right)^{-1}d\theta^2 \;+\; \rho(\theta)^2\,d\Omega^2\,.} \tag{33}$$

This is the static, spherically symmetric vacuum solution of the Einstein field equations in the phase parameterisation.

5 Curvature and Singularity Analysis

5.1 The Kretschmann scalar

For the metric (33), the Riemann tensor has the same nonzero components as the standard Schwarzschild metric, expressed in terms of the areal radius. The Kretschmann scalar evaluates to

$$K(\theta) \;=\; \frac{48\,G^2M^2}{\rho(\theta)^6}\,. \tag{34}$$

5.2 Boundedness on the physical domain

Proposition 5 (Curvature bound)**.** *For all $\theta \in [0,\infty)$, the Kretschmann scalar* (34) *satisfies*

$$0 \;<\; K(\theta) \;\le\; 48\,G^2M^2\,, \tag{35}$$

with the maximum attained at $\theta = 0$, where $\rho(0) = 1$, and with $K(\theta) \to 0$ as $\theta \to \infty$.

Proof. By Lemma 2(v), $\rho(\theta) \ge 1$ for all $\theta \in [0,\infty)$, so $\rho(\theta)^6 \ge 1$, and hence $K(\theta) \le 48G^2M^2$. The maximum is attained at $\theta = 0$ where $\rho = 1$. By Lemma 2(iii), $\rho(\theta) \to \infty$, so $K(\theta) \to 0$. Positivity is immediate from (34) and $\rho > 0$. □

Corollary 6 (No curvature singularity)**.** *The metric* (33) *is geodesically complete in the radial direction and free of curvature singularities on its domain.*

Proof. By Proposition 5, K is bounded on $[0,\infty)$. All other algebraic curvature invariants R, $R_{\mu\nu}R^{\mu\nu}$, etc., are likewise bounded (in vacuum, $R = 0$ and $R_{\mu\nu} = 0$ identically, so these invariants vanish; only Weyl-tensor invariants contribute, and they are bounded together with K). The metric coefficients $F(\theta)$ and $G(\theta)$ are smooth and positive on the physical domain unless a horizon is present (see below), so no metric singularity arises. □

5.3 Horizon analysis

The coefficient $F(\theta) = 1 - 2GM/\rho(\theta)$ vanishes when $\rho(\theta) = 2GM$, i.e. when

$$1 + \frac{2\theta_h}{\Theta_{\max}} = 2GM \quad\Longleftrightarrow\quad \theta_h = \frac{\Theta_{\max}(2GM-1)}{2}\,. \tag{36}$$

Two cases arise:

(a) If $2GM < 1$ (in the chosen units), then $\theta_h < 0$, which lies outside the physical domain $[0,\infty)$. No horizon is present.

(b) If $2GM \ge 1$, then $\theta_h \ge 0$ lies within the physical domain, and a horizon is present at θ_h. This horizon is a coordinate singularity of the chart, removable by transition to Kruskal-Szekeres-type coordinates in the standard manner.

In both cases, the geometry behind the horizon (where present) extends smoothly to all $\theta \in [0, \theta_h)$, with curvature bounded by $48G^2M^2$. No curvature divergence is present at any finite θ.

6 Recovery of the Standard Form

We now establish that the metric (33) reproduces the standard Schwarzschild metric (1) under change of radial coordinate.

Theorem 7 (Equivalence to Schwarzschild)**.** *Under the change of coordinate $r = \rho(\theta)$, the metric* (33) *on the chart $\theta \in [0, \infty)$ transforms into the Schwarzschild metric* (1) *on the chart $r \in [1, \infty)$.*

Proof. From Definition 1, $r = \rho(\theta) = 1 + 2\theta/\Theta_{\max}$, so

$$dr \;=\; \rho'(\theta)\, d\theta \;=\; \frac{2}{\Theta_{\max}}\, d\theta\,, \qquad \text{equivalently} \qquad d\theta \;=\; \frac{\Theta_{\max}}{2}\, dr\,. \tag{37}$$

The $d\theta^2$ term in (33) becomes

$$\left(\frac{2}{\Theta_{\max}}\right)^2 d\theta^2 \;=\; \left(\frac{2}{\Theta_{\max}}\right)^2 \cdot \left(\frac{\Theta_{\max}}{2}\right)^2 dr^2 \;=\; dr^2\,. \tag{38}$$

The factors of $2/\Theta_{\max}$ cancel exactly. Substituting into (33):

$$ds^2 \;=\; -\left(1 - \frac{2GM}{r}\right) dt^2 \;+\; \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 \;+\; r^2\, d\Omega^2\,, \tag{39}$$

on the domain $r \in [1, \infty)$. This is precisely (1) restricted to $r \geq 1$. □

Corollary 8 (Identity of physical predictions)**.** *On the domain $r \in [1, \infty)$, the metric* (33) *predicts identical observational consequences to the standard Schwarzschild metric* (1)*. In particular: light deflection, perihelion precession, Shapiro time delay, gravitational redshift, and (where $2GM \geq 1$) the existence and properties of the event horizon are reproduced exactly.*

7 Status of the Locus $r = 0$

7.1 The locus is not in the domain

The Schwarzschild metric (1) in standard form is conventionally taken to be defined on the domain $r \in (0, \infty)$, with a curvature singularity at $r = 0$. In the phase parameterisation of the present derivation, the equivalent statement is that the metric (33) is defined on $\theta \in [0, \infty)$, and $r = 0$ corresponds via $r = \rho(\theta) = 1 + 2\theta/\Theta_{\max}$ to

$$\theta_* = -\frac{\Theta_{\max}}{2}\,, \tag{40}$$

which lies outside the physical domain.

7.2 The singularity is an artefact of analytic continuation

The Kretschmann scalar (34) is finite for all $\theta \in [0, \infty)$ and bounded by $48G^2M^2$ on this domain. Its formal extension to $\theta \in (-\Theta_{\max}/2, 0)$ would produce values of $\rho(\theta) \in (0, 1)$ and corresponding values of $K(\theta) \in (48G^2M^2, \infty)$, with $K \to \infty$ as $\theta \to -\Theta_{\max}/2^+$.

However, this extension is not part of the physical solution. The phase parameter θ has domain $[0, \infty)$ by Definition 1, and values $\theta < 0$ are not in the domain. They correspond to a hypothetical evolution of the phase parameter *prior to its origin*, which is excluded by the construction.

Remark 9 (Nature of the singularity)**.** The Schwarzschild singularity at $r = 0$ is therefore a singularity of the *analytic continuation* of the metric to values of the radial coordinate that the radial parameter does not reach. It is not a singularity of the geometry; it is a singularity in the chart that arises when the chart is extended past the image of the radial parameter's domain.

7.3 Implications for the singularity theorems

The singularity theorems of Penrose [2] and Hawking–Penrose [3], when applied to Schwarzschild geometry, conclude that geodesics terminate in finite affine parameter on a curvature singularity. The premises of these theorems include conditions on the global structure of the manifold and on the matter content. Under those premises, the theorems are correct.

The present derivation does not contradict the singularity theorems. It shows that the premise of the theorems—in particular, the choice of parameterising the radial direction with a coordinate r that ranges over $(0, \infty)$—is one possible choice, and that there exist alternative parameterisations in which the radial coordinate is the image of a phase parameter with a domain bounded away from the singular locus. Under such parameterisations, the singularity locus is not in the domain of the manifold, and the conclusion of the singularity theorems does not apply, because the manifold whose existence and structure they assume is not the manifold under consideration.

8 Conclusion

The Schwarzschild solution to the Einstein vacuum equations, in its standard form (1), is defined on the domain $r \in (0, \infty)$ and exhibits a curvature singularity at $r = 0$. When the radial direction is parameterised instead by a phase parameter $\theta \in [0, \infty)$ via a generative function $\rho(\theta) = 1 + 2\theta/\Theta_{\max}$ with image $[1, \infty)$, the same Einstein equations are solved, the same exterior geometry is obtained, the same observational consequences are predicted, and the locus $r = 0$ does not appear in the physical domain.

The Schwarzschild singularity is therefore not a feature of the geometry. It is a feature of the choice to parameterise the radial direction by a coordinate r that is treated as primitive, with domain $(0, \infty)$, rather than as a derived quantity with domain determined by the generative function from which it is constructed. The derivation requires no modification of the field equations, no exotic matter, no quantum corrections, and no exotic geometry. It requires only the recognition that the radial coordinate is, in any physical setting, derived from a more primitive geometric quantity, and that the domain of the derived coordinate is determined by the domain of the primitive parameter.

Coda

A century of theoretical physics has been spent investigating the structure of the locus $r = 0$ in the Schwarzschild solution: the formation of singularities under gravitational collapse, their concealment behind event horizons, the loss or preservation of information upon their formation, the possible quantum-gravitational resolution of their divergent curvature, the cosmological extension of the singularity to the origin of time, and the entire industry of speculative interior structure. Books have been written, careers built, prizes awarded.

The locus in question is, by stipulation, a single point in the underlying manifold: zero-dimensional, dimensionless, possessing no extent in any direction. The question of its geometric properties, on reflection, admits a simple and complete answer.

It has none.

It is a point.

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